

Do Simple Models Still Win? Extending the LTSF-Linear Benchmark to Short-Horizon Sports Forecasting

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Introduction

Time series analysis involves the extraction of meaningful patterns from historical data series, and within the domain of college football, this presents a compelling opportunity to assess how team performance evolves over the course of a season. Sports science research teams are being developed at increasing rates as data mining has matured, and research groups have begun to harness game-level data to generate predictions and actionable insights that can drive team performance and inform strategic decision making. Exploring this data lays the groundwork for how time series analysis can shape future methods in college football analytics.

Despite significant work at the professional level, the collegiate space remains relatively underexplored. College football presents unique challenges that distinguish it from professional leagues as roster turnover is affected by the transfer portal, season-ending injuries, and graduating players entering the workforce or professional play. Much of the existing work focuses on team-level performance as a whole, as granular player-level data is often inaccessible across public databases. There is also a notable gap in the literature around the prediction of home spread, defined here as the point margin of victory by the home team where a positive value indicates a win and a negative value indicates a loss. Most spread prediction work relies on expert opinion, and while data mining has begun to enter this space, the use of sequential game-level statistics to identify patterns in spread across a season remains limited.

In the broader landscape of time series forecasting algorithms, Transformer-based models have dominated recent research, but Zeng et al. [1] challenged this trend by demonstrating that the self-attention mechanism inherent to Transformers hinders their effectiveness with time series data. They proposed a family of simple linear models, denoted as LTSF-Linear, that outperform Transformer architectures on standard benchmarks using look-back windows ranging from 24 to 720 timesteps. However, college football home game seasons are naturally short, often capped at around seven games, which falls well outside the window sizes examined in the original work. This project aims to assess whether the same finding of simpler models outperforming more complex architectures holds under shorter time series windows, and whether it extends to a noisier target variable with less inherent trend and periodicity than the benchmark datasets used in the paper. The contributions of this project are implementing the DLinear variant of LTSF-Linear alongside an LSTM and a vanilla Transformer for comparison, validating model performance on a football domain dataset using home game statistics from Football Bowl Subdivision (FBS) college football teams spanning 2014 to 2024, and testing the generalizability of the original paper’s findings to shorter sequence lengths and a noisier prediction target.

Related Work

A major direction in sports outcome prediction has focused on feature-based machine learning models that rely on engineered attributes derived from raw game statistics. For example, prior work by Leung and Joseph [2] developed a custom algorithm using derived features such as rush-pass ratios to predict bowl game outcomes, demonstrating that transforming raw statistics into more meaningful representations can

significantly improve predictive performance. Similarly, Gifford and Bayrak [3] showed that tree-based models such as decision trees and random forests achieved around 80% accuracy in predicting NFL game outcomes. While effective, these approaches largely treat each game as an independent instance and do not explicitly model temporal dependencies across games. A second direction focuses on sequential modeling approaches that explicitly account for temporal structure in the data. In particular, Long Short-Term Memory (LSTM) networks [4] have been widely used for time series forecasting due to their ability to capture long-term dependencies through gated memory mechanisms. These models allow past observations to influence future predictions in a structured way, making them well-suited for sequential data such as game-by-game team performance. However, LSTMs can be computationally expensive and may struggle when data is limited or when very long-range dependencies must be captured.

More recently, Transformer-based architectures have emerged as an alternative to recurrent neural networks for time series forecasting. Transformers rely on self-attention mechanisms [5] to model relationships between all elements in a sequence, enabling them to capture global dependencies more efficiently than LSTMs. However, because self-attention is permutation invariant, additional mechanisms such as positional encodings are required to preserve temporal ordering. This has led to a variety of adaptations aimed at improving their effectiveness for time series forecasting, reflecting an ongoing effort to balance model complexity with the need to accurately represent temporal structure. Across these directions, a key distinction lies in how temporal information is handled: feature-based models largely ignore it, recurrent models like LSTMs encode it sequentially, and Transformer-based models attempt to learn it through attention mechanisms. The current work builds on this landscape by examining how model complexity and temporal representation impact predictive performance in the context of sports data, particularly under constraints such as shorter sequence lengths.

Current Work

The paper that is used as a basis of study is titled Are Transformers Effective For Time Series Forecasting [1], and proposes an ‘embarrassingly simple’ baseline algorithm that can be used in future tests of the long-term time series forecasting (LTSF) problem. The paper serves as an empirical study on the effectiveness of a more complex model that has inherent flaws in this kind of problem in comparison to simple models that can outperform these models and is a direct application of Occam’s razor with relevant datasets. In the paper, Zeng et al. talk about the multitude of Transformer-based architectures that are being used to solve problems in this area, and that with these problems the temporal ordering is the most important part of the data that is being mined/assessed. Transformer models use self-attention, which is a type of architecture that allows data points to be learned in context to one another, but self-attention is permutation invariant, meaning that order is inherently ignored by the process. Further iterations of the Transformer model have introduced different types of embeddings to mitigate the issue: simple additions of positional encodings, learnable temporal embeddings (Autoformer, Informer) that allow the model to learn the embeddings instead of ignoring them, logsparse masking (LogTrans) that allow the model to densely attend to recent data and sparsely to less recent data, and frequency domain processing (FEDformer) where the sequence is encoded into a frequency domain before applying attention so the temporal information is baked into the attention process. While all of these have made improvements in the effectiveness of Transformers in this space, the authors questioned whether the self-attention was still getting in the way of the performance of these models and if the fixes were only superficial in nature and thus proposed a set of linear models called LTSF-linear (essentially three different linear regression models) that outperformed the Transformer models on 9 different benchmark datasets within LTSF problems.

They define long-term time series problems as follows: For time series containing C variates, given historical data $X = \{X_t^1, \dots, X_t^C\}_{t=1}^L$, wherein L is the look-back window size and X_t^i is the value of the i th variate at the t th time step. The time series forecasting task is to predict the values $\hat{X} = \{\hat{X}_t^1, \dots, \hat{X}_t^C\}_{t=L+1}^{L+T}$ at the T future time steps. Within these types of problems, prediction can use Iterative Multi-step Forecasting (IMS), which focuses on a single-step forecast in which each prediction is used to inform the next, or can use Direct Multi-step Forecasting (DMS), which uses the full history of the look-back window to inform all future predictions. In previous comparisons of the effectiveness of Transformers in relation to simpler models, Transformers were using DMS to inform predictions, while the other models were using IMS; because

IMS uses each prediction to inform the next, error accumulation can become a problem where future time steps have a worse chance of being predicted correctly, whereas DMS does not suffer from the same error accumulation. The authors proposed that this mismatch in prediction strategy may have inherently boosted the performance of Transformers, and all of their experiments with LTSF-Linear used DMS to make the comparison more accurate.

For LSTF-Linear, the base formula uses linear regression through time series, where the mathematical expression is defined as $\hat{X}_i = WX_i$ where $W \in \mathbb{R}^{T \times L}$ is a linear layer along the temporal axis, and $\hat{X}_i$ and X_i are the prediction and input for each i th variate. Their vanilla Linear model essentially follows standard linear regression, and they adapted two different models denoted as DLinear and NLinear. DLinear utilizes the decomposition scheme used in Autoformer/FEDformer in which the raw data is split into a trend component X_{trend} that uses a moving average kernel to form the data points, and a seasonal (remainder) component $X_{\text{remainder}} = X - X_{\text{trend}}$ that comes from the subtraction of the trend from the raw data and both sets are run through separate linear models W_{trend} and $W_{\text{remainder}}$, and the outputs are summed to inform a final prediction $\hat{X}_i = W_{\text{trend}}X_{\text{trend}} + W_{\text{remainder}}X_{\text{remainder}}$; this builds on the performance of vanilla Linear when there is a clear trend in the data since the decomposition explicitly handles trend (and is what was chosen to be implemented in this application of LTSF-Linear). NLinear builds on vanilla linear when there is a clear distribution shift in the dataset, in which NLinear first computes $X' = X - x_L$ where x_L is the last value in the input sequence, passes X' through a linear layer, then adds x_L back to the final prediction as $\hat{X} = WX' + x_L$, effectively normalizing the input around the most recent observation.

The paper, through many different experiments, found that Transformers were being outperformed by this set of linear models in all 9 of the benchmark datasets used, and did a few studies hinting at the use of self-attention being one of the main cruxes of the capabilities of Transformers in these problems, and urged future research to look into new algorithms (including new Transformer models that can outperform LTSF-Linear). They looked at longer sequences of data in their work, and for this data the project is looking at the home game seasons of teams in the FBS, and testing the similarities in performance differences as it pertains to shorter sequences that their paper highlights.

Experimental Evaluation

Dataset

As mentioned before, this project’s main goals are to implement the DLinear variant of LTSF-Linear alongside an LSTM and a vanilla Transformer for comparison, validate the model’s performances on a football domain dataset comprising home game statistics from FBS college football teams spanning 2014 to 2024, and test the generalization of the original paper’s findings to shorter sequence lengths and a noisier prediction target. In the experimental setup, the dataset used is a feature reduced set that originates from the Kaggle dataset College Football Game Stats, 2002 to January 2026. The dataset is composed of box game scores/game statistics from various college teams that range from 2002 to January 2026, though at the time of initial analysis the dataset only comprised games from 2002 to January 2025. The set includes attributes such as season, number of first downs in various quarters, and team ranks, to name a few. To narrow the dataset to a meaningful number of games, the dataset was reduced to looking at two conferences, the Big Ten Conference and the Southeastern Conference, and reduced further to look at their home games only from 2002 to 2024 in order to facilitate the look back windows in question. After consulting related work and assessing different statistics that are derived and used in real-world football analysis, a set of derived features were created and used in the experiments; Table 1 provides a list of features utilized as well as their categorizations by meaning of usage. The reduced data set yielded 2,630 games, and to split the data into training, validation, and test, the training data fell from 2002 - 2022, validation in the 2023 season, and test in the 2024 season.

Feature Group	Variables
Game Context	week, home_season, game_type, neutral
Team Strength	rank_home, rank_away
Scoring Performance	q1_home, q2_home, q3_home, q4_home
Offensive Production	first_downs_home, possession_home, Run-Pass Ratio, Rush Yards - Total Yards
Turnovers	fum_home, int_home
Penalties	pen_num_home, pen_yards_home
Efficiency Metrics	Home Third Down Completion Rate, Home Pass Completion Rate
Target Variable	spread

Table 1: Feature Groups Used for Game Spread Prediction

Methodology

Each training sample is constructed as a sliding window of W past games, formally represented as $X_{t-W:t} \in \mathbb{R}^{W \times F}$, where $W \in \{3, 4, 5\}$ denotes the number of past games in the window and F denotes the number of engineered game-level features. The sequences used are constructed independently per team to preserve their temporal ordering and to make sure there is no leakage between different teams. The prediction target for each sample is the point spread of the next game, $y_{t+1} \in \mathbb{R}$, and each model produces a direct estimate $\hat{y}_{t+1}$ in a single forward pass. All models follow a direct single-step forecasting strategy, $X_{t-W:t} \rightarrow \hat{y}_{t+1}$, with no iterative or autoregressive prediction used at any stage.

Three separate models were evaluated in this project; DLinear decomposes the input sequence into a trend component and a seasonal component using a moving average kernel, trains two separate linear regression models on each component independently, and produces a final prediction by summing both outputs. The LSTM used here is a sequential model with a hidden memory state of 64 dimensions and a single-layer architecture, allowing past game observations to influence future predictions through its gated memory mechanism. The vanilla Transformer used here projects each timestep into a 64-dimensional space, uses a single Transformer encoder layer with 4 attention heads, and applies learned positional encodings to inject temporal ordering into the input sequence. The LSTM and vanilla Transformer model were modified using existing architecture in Pytorch as a basis.

Each model was trained and evaluated under a unified hyperparameter search across the following configurations:

- **Window Size:** 3, 4, 5
- **Learning Rate:** 0.0001, 0.0005, 0.001, 0.005
- **Epochs:** 50, 100, 150, 200
- **L1 Regularization:** None, 0.0001, 0.001
- **L2 Regularization:** None, 0.0001, 0.001

This yielded 432 total hyperparameter configurations per model. All models were trained on the same dataset split and evaluated using the same metrics as the original paper: Mean Squared Error (MSE) and Mean Absolute Error (MAE). For reporting purposes, only the top 10 performing configurations per model are presented in the results, though metrics were recorded for all 432 configurations in the code.

Results

Contrary to the findings of Zeng et al., the Transformer outperformed both the LSTM and DLinear on this dataset, suggesting that the relationship between model complexity and predictive performance may shift when the target variable is noisier and exhibits less linear trend than the benchmark datasets examined in the original paper. The improvement is not dramatic, but it is consistent, and the results reveal a natural ordering in model performance where complexity corresponds to marginal but measurable gains in both MSE

and MAE (see Table 3). This also reinforces the utility of DLinear as a baseline, since even when it does not win, it serves as a meaningful reference point for gauging how much more complex models are actually contributing. Table 2 below shows the findings of MSE and MAE with each model, where we see the natural ordering of better performance with increasing complexity.

Model	MSE	MAE
DLinear	400.1035	15.8810
LSTM	394.8339	15.5615
Transformer	381.8938	15.2283

Table 2: Best Achieved MSE and MAE Per Model

Across the chosen hyperparameter configurations, several patterns emerged that seemed interesting. Longer window sizes of 4 and 5 consistently outperformed a window of 3, which aligns with the intuition that more historical context generally supports better prediction; further studies could look into assessing the entire season rather than just home games to see if this increase in window size extends. Smaller learning rates in the range of 0.0001 to 0.0005 appear to be a meaningful driver of performance for DLinear and the Transformer in particular, while the LSTM and Transformer both tended to perform better with fewer training epochs, suggesting that these models reach their optimal generalization point earlier and begin to overfit with continued training. L1 and L2 regularization did not produce a clear or consistent effect across configurations, with the best performing combinations appearing somewhat sporadically across the regularization grid; this was interesting since the other hyperparameters of window, epoch size, and learning rate seemed to have more clear patterns, and suggest that L1 and L2 regularization don’t have much of an effect on noisier data when it comes to weight configuration.

Rank	DLinear								LSTM								Transformer							
	W	LR	E	L1	L2	MSE	RMSE	MAE	W	LR	E	L1	L2	MSE	RMSE	MAE	W	LR	E	L1	L2	MSE	RMSE	MAE
1	5	0.0001	200	0.001	0.0	400.10	20.00	15.88	5	0.005	100	0.0	0.0001	394.83	19.87	15.56	4	0.0005	100	0.001	0.0	381.89	19.54	15.23
2	4	0.0005	100	0.0001	0.0001	400.83	20.02	15.96	5	0.005	100	0.0	0.001	396.11	19.90	15.77	5	0.001	100	0.001	0.001	386.17	19.65	15.55
3	5	0.0001	150	0.001	0.0	401.21	20.03	15.87	5	0.005	200	0.0	0.001	398.73	19.97	15.68	4	0.0005	150	0.0001	0.001	386.48	19.66	15.29
4	5	0.001	100	0.001	0.001	401.45	20.04	15.52	5	0.005	50	0.0001	0.0	400.20	20.01	15.69	5	0.0005	100	0.0001	0.0	388.71	19.72	15.61
5	4	0.001	50	0.0	0.0	401.48	20.04	15.87	5	0.005	150	0.001	0.0	400.31	20.01	15.65	5	0.0005	100	0.001	0.0	390.66	19.77	15.35
6	4	0.0001	50	0.0	0.0001	401.64	20.04	15.79	5	0.001	50	0.0001	0.001	400.58	20.01	15.93	4	0.0005	200	0.001	0.001	390.85	19.78	15.49
7	4	0.001	200	0.0001	0.001	402.27	20.06	15.89	4	0.0001	100	0.0001	0.001	400.91	20.02	15.74	4	0.0005	100	0.0001	0.001	391.82	19.80	15.63
8	5	0.005	200	0.0	0.0	402.77	20.07	15.80	4	0.0001	50	0.001	0.001	402.64	20.07	15.77	5	0.005	200	0.0001	0.0001	392.76	19.82	15.83
9	4	0.001	50	0.001	0.0	402.89	20.07	15.94	4	0.0005	150	0.0	0.0	402.99	20.07	15.72	4	0.001	200	0.001	0.0	394.09	19.85	15.70
10	4	0.001	200	0.001	0.0001	402.92	20.07	15.97	4	0.001	150	0.0	0.0001	403.56	20.09	15.74	4	0.0005	100	0.0	0.0001	394.32	19.86	15.43

Table 3: Top 10 Configurations Per Model (Sorted by MSE Rank)

Overall, the model type appears to be the primary driver of predictive performance rather than specific hyperparameter tuning, as the gains between configurations within each model are relatively small. This stability across configurations reflects the inherent noisiness of point spread data, which all three models approximate with comparable difficulty. DLinear remains a strong and interpretable baseline, and the results suggest that while complexity offers some advantage in this setting, the margin is modest enough that simpler approaches remain competitive.

Contributions

I contributed to the completion of the project alone. :)

References

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